

PHYS 380 - Home work 1

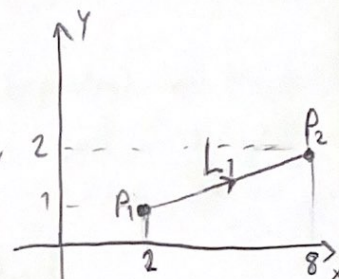
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2022 401 207

1. Given $\vec{F} = a_x y + a_y x$, find $\int_L \vec{F} \cdot d\vec{l}$
from $P_1(2,1,-1)$ to $P_2(8,2,-1)$

a) along a straight line joining the two point

• Let us call this path L_1 . L_1 can be parametrized by

$$x = x, \quad y = \frac{1}{6}x + \frac{2}{3}, \quad z = -1 \Rightarrow dy = \frac{dx}{6}$$



$$\rightarrow d\vec{l} = dx \vec{a}_x + dy \vec{a}_y = dx \vec{a}_x + \frac{1}{6} dx \vec{a}_y$$

$$\rightarrow \vec{F} \cdot d\vec{l} = \left(a_x \cdot \frac{x+4}{6} + a_y x \right) \cdot \left(a_x dx + \frac{1}{6} a_y dx \right) = \frac{x+4}{6} dx + \frac{x}{6} dx = \frac{x+2}{3} dx$$

$$\rightarrow \int_{L_1} \vec{F} \cdot d\vec{l} = \int_2^8 \frac{x+2}{3} dx = \frac{1}{3} \left[\frac{x^2}{2} + 2x \right]_2^8 = \frac{1}{3} (30 + 12) = \boxed{14}$$

b) along the parabola $x = 2y^2$

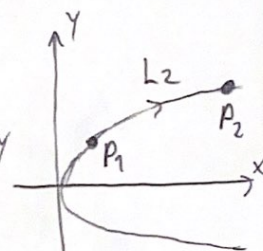
• Let us call this path L_2 . L_2 can be parametrized by

$$x = 2y^2, \quad y = y, \quad z = -1 \Rightarrow dx = 4y dy$$

$$\rightarrow d\vec{l} = dx \vec{a}_x + dy \vec{a}_y = 4y dy \vec{a}_x + dy \vec{a}_y$$

$$\rightarrow \vec{F} \cdot d\vec{l} = (a_x y + a_y 2y^2) \cdot (a_x 4y dy + a_y dy) = 4y^2 dy + 2y^2 dy = 6y^2 dy$$

$$\rightarrow \int_{L_2} \vec{F} \cdot d\vec{l} = \int_1^2 6y^2 dy = 6 \left[\frac{y^3}{3} \right]_1^2 = 2 \cdot 7 = \boxed{14}$$



• An important property of conservative fields is path independence meaning that the line integral along any two arbitrary points is independent of the path followed. We see that both paths resulted in the same number which was necessary for $\vec{F}$ to be a conservative field. Regardless, it's not sufficient.

Assume that vector field $\vec{F}$ is the gradient of some scalar field.

$$\vec{F} = \nabla \phi$$

Then by the gradient theorem, line integral of $\vec{F}$ along a path C is

$$\int_C \vec{F} \cdot d\vec{\ell} = \int_C \nabla \phi \cdot d\vec{\ell} = \phi(B) - \phi(A)$$

, where A, B are the endpoints of C , which is independent of the path.

Then we say that $\vec{F}$ is a conservative field. Since curl of a gradient is always zero, we can state:

$$\nabla \times (\nabla \phi) = \nabla \times \vec{F} = 0 \quad \Rightarrow \text{for conservative fields.}$$

Checking the curl of $\vec{F} = y a_x + x a_y$,

$$\nabla \times \vec{F} = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & x & 0 \end{vmatrix} = -a_x \underbrace{\frac{\partial x}{\partial z}}_0 + a_y \underbrace{\frac{\partial y}{\partial z}}_0 + a_z \underbrace{\left(\frac{\partial x}{\partial x} - \frac{\partial y}{\partial y} \right)}_0 = 0$$

since x, y, z are independent

We see that the field $\vec{F} = \underline{a_x y} + \underline{a_y x}$ is conservative.

2. Find the divergence of radial fields

$$a) f_1(\vec{R}) = \hat{a}_R R^n$$

$$b) f_2(\vec{R}) = \hat{a}_R \frac{K}{R^2}$$

We shall first derive the divergence for spherical coordinates using

$$\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 A_{u_1}) + \frac{\partial}{\partial u_2} (h_1 h_3 A_{u_2}) + \frac{\partial}{\partial u_3} (h_1 h_2 A_{u_3}) \right]$$

$$\left. \begin{array}{l} u_1 = R, \quad h_1 = 1 \\ u_2 = \theta, \quad h_2 = R \\ u_3 = \phi, \quad h_3 = R \sin \theta \end{array} \right\} \nabla \cdot \vec{A} = \frac{1}{R^2 \sin \theta} \left[\frac{\partial}{\partial R} (R^2 \sin \theta A_R) + \frac{\partial}{\partial \theta} (R \sin \theta A_\theta) + \frac{\partial}{\partial \phi} (R A_\phi) \right]$$

$$\nabla \cdot \vec{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$a) f_1(\vec{R}) = \hat{a}_R R^n \Rightarrow A_R = R^n, A_\theta = 0, A_\phi = 0$$

$$\nabla \cdot f_1(\vec{R}) = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 R^n) = \frac{(n+2) R^{n+1}}{R^2} = \boxed{(n+2) R^{n-1}}$$

$$b) f_2(\vec{R}) = \hat{a}_R \frac{k}{R^2} \Rightarrow A_R = \frac{k}{R^2}, A_\theta = 0, A_\phi = 0$$

$$\nabla \cdot f_2(\vec{R}) = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{k}{R^2} \right) = \frac{1}{R^2} \frac{\partial k}{\partial R} = \boxed{0} \quad \text{since } k \text{ is constant.}$$

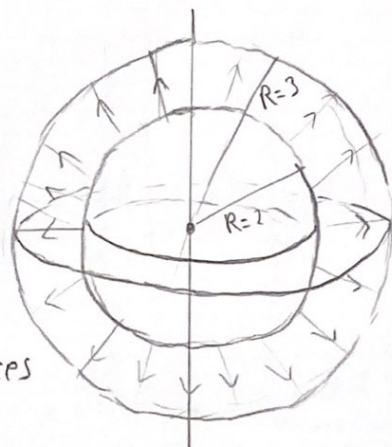
3. A vector field $\vec{D} = \hat{a}_R (\cos^2 \phi) R^2$ exist in the region between two spherical shells defined by $R=2$ and $R=3$. Evaluate

$$a) \oint \vec{D} \cdot d\vec{S}$$

For spherical coordinates:

$$dS = \hat{a}_R R^2 \sin \theta d\theta d\phi$$

where $\hat{a}_R$ is the unit vector pointing radially outwards. We integrate over two spherical surfaces $S_1 (R=R_1=2)$ and $S_2 (R=R_2=3)$.



$$\vec{dS}_1 = -\hat{a}_R R_1^2 \sin \theta d\theta d\phi \Rightarrow "-" \text{ sign since inward boundary}$$

$$\vec{dS}_2 = +\hat{a}_R R_2^2 \sin \theta d\theta d\phi \Rightarrow "+" \text{ sign since outward boundary}$$

$$\oint_{S=S_1+S_2} \vec{D} \cdot d\vec{S} = \oint_{S_1} \vec{D} \cdot d\vec{S}_1 + \oint_{S_2} \vec{D} \cdot d\vec{S}_2 = \oint_{S_1} \cos^2 \phi R_1^2 (-R_1^2 \sin \theta d\theta d\phi) + \oint_{S_2} \cos^2 \phi R_2^2 (+R_2^2 \sin \theta d\theta d\phi)$$

$$= \int_0^{2\pi} \int_0^\pi (R_2^4 - R_1^4) \sin \theta \cos^2 \phi d\theta d\phi = (R_2^4 - R_1^4) \left[\int_0^\pi \sin \theta d\theta \right] \left[\int_0^{2\pi} \cos^2 \phi d\phi \right]$$

$\hookrightarrow$ no R integration
since it's constant

$$\oint \vec{D} \cdot d\vec{S} = (R_2^4 - R_1^4) \left(\int_0^\pi \sin\theta d\theta \right) \left(\int_0^{2\pi} \underbrace{\cos^2\phi d\phi}_{\frac{1+\cos 2\phi}{2}} \right)$$

$$= (3^4 - 2^4) \left[\cos\theta \right]_\pi^0 \left[\frac{\phi}{2} + \frac{\sin 2\phi}{4} \right]_0^{2\pi} = 65 \cdot 2 \cdot \pi = \boxed{130\pi}$$

b) $\int \nabla \cdot \vec{D} \, dV$

We've found in question 2 that divergence $\nabla \cdot \vec{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_r) + \frac{1}{R \sin\theta} \frac{\partial}{\partial \theta} (R \sin\theta A_\theta) + \frac{1}{R \sin\theta} \frac{\partial A_\phi}{\partial \phi}$ for spherical coordinates is :

Then $\nabla \cdot \vec{D} = \nabla \cdot (\hat{a}_R \cos^2\phi R^2) = \frac{1}{R^2} \frac{\partial}{\partial R} (R^4 \cos^2\phi) = 4R \cos^2\phi$

$\rightarrow dV = R^2 \sin\theta \, d\theta \, d\phi \, dR$

$$\int_V \nabla \cdot \vec{D} \, dV = \int_V (4R \cos^2\phi) R^2 \sin\theta \, d\theta \, d\phi \, dR = \int_0^{2\pi} \left[\int_0^\pi \left[\int_2^3 4R^3 dR \right] \sin\theta \, d\theta \right] \cos^2\phi \, d\phi$$

$$= \left[\int_0^{2\pi} \cos^2\phi \, d\phi \right] \left[\int_0^\pi \sin\theta \, d\theta \right] \left[\int_2^3 4R^3 dR \right] = \pi \cdot 2 \cdot (3^4 - 2^4) = \boxed{130\pi}$$

$\rightarrow$ It is not a coincidence that two integrals gave the same result. By divergence theorem, surface integral of a vector field over a closed surface S is equal to volume integral of the divergence of the field over the volume S encloses.

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V \nabla \cdot \vec{D} \, dV$$